

An Indoor Environmental Monitoring System Using an Ontology-Based Context and Rule-Based Controlled LLM Framework

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Abstract

As the importance of managing indoor environmental comfort, safety, and energy efficiency has increased, growing attention has been paid to technologies that can reliably interpret real-time environmental conditions and deliver them to users. However, large language models (LLMs) may generate hallucinations when explaining environmental states, such as information that does not match actual sensor data or exaggerated descriptions, which can reduce the reliability of indoor environmental monitoring systems. This study proposes an indoor environmental monitoring framework that integrates ontology-based semantic context and grounding structures, rule-based state assessment, and a controlled LLM based on indoor environmental sensor data.

The experimental results show that the proposed structure achieved higher state consistency and explanatory reliability than a general LLM. In addition, a function for controlling air-conditioner power based on occupancy was implemented, and the study suggests the potential for future expansion toward ventilation control and detailed HVAC control.

Keywords: Ontology-based context, controlled LLM, rule-based state assessment, hallucination mitigation, indoor environmental monitoring, occupancy detection

1. Introduction

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In multi-use facilities such as schools, hospitals, and offices, the indoor environment has a significant impact on user health and safety. In particular, environmental factors such as CO₂, PM_{2.5} (fine particulate matter), temperature, and humidity are closely associated with reduced concentration and respiratory diseases, and their effects become more pronounced in enclosed spaces. Accordingly, active research has been conducted on monitoring indoor environmental conditions in real time and presenting them to users in an intuitive manner. In addition, in multi-use spaces such as educational facilities, unnecessary energy consumption may occur when heating and cooling systems are operated without considering occupancy. Therefore, there is increasing demand for intelligent environmental management systems that can support efficient indoor environmental management and energy saving by jointly utilizing real-time environmental conditions and occupancy information.

Conventional indoor environmental monitoring systems primarily provide sensor data in numerical or graphical form. However, this approach has the limitation that users may find it difficult to intuitively understand the current environmental state. Recently, studies using large language models (LLMs) to explain sensor data in natural language have increased; however, such approaches may cause hallucinations, in which false information not grounded in the input data is generated. In fields where reliability is essential, such as environmental monitoring, the provision of inaccurate information can critically affect user decision-making.

Ontology has been used as a knowledge representation structure that can semantically connect diverse sensor data and spatial information. Rule-based reasoning can also provide explicit criteria and explainability, and thus offers high reliability in safety-related domains. However, existing studies have mainly focused on ontology-based data structuring or rule-based assessment, and research linking these approaches with LLM-based natural language generation remains insufficient.

Therefore, this study proposes a controlled LLM-based indoor environmental monitoring framework that combines an ontology-based context structure with rule-based state assessment. To this end, CO₂, PM_{2.5}, temperature, humidity, and occupancy data were collected using an Environmental Data Collector (EDC), an Infrared Camera (IRC), and a Radar Detector (RDC), and the collected data were represented in an ontology structure. Natural language explanations were then generated based on the results of rule-based state assessment to improve explanatory reliability.

The main contributions of this study are as follows.

First, heterogeneous indoor environmental sensor data were integrated into an ontology-based context structure.

Second, the reliability of explanations was improved by combining rule-based state assessment with a controlled LLM structure.

Third, hallucination was minimized by applying a state-based output restriction structure.

Fourth, the practical applicability of the proposed approach was confirmed by linking the generated results with a real-time dashboard and TTS functionality.

2. Related Work

2.1 Indoor Environmental Monitoring Studies

Indoor environmental monitoring is a technology for measuring environmental factors such as temperature, humidity, CO₂, and PM_{2.5} in real time and analyzing the environmental conditions of a space. Recently, studies have been conducted to collect indoor air quality data using integrated environmental sensors and

various IoT devices and to analyze environmental changes by space [1]. Research has also been conducted on multi-sensor-based integrated indoor air quality monitoring architectures to more accurately identify air quality conditions by space [2].

2.2 Occupancy Detection and Multi-Sensor-Based Environmental Analysis Studies

Occupancy information is used as key information for reducing unnecessary heating, cooling, and power consumption and for efficiently managing environmental conditions by space. Kim proposed a system that remotely and automatically controls legacy air conditioners without IoT functionality using motion detection sensors and temperature-humidity sensors, reducing power consumption by automatically stopping the air conditioner when the space is vacant and maintaining an appropriate temperature [3].

Cui et al. proposed an occupancy detection method using low-resolution thermal images and a YOLOv5-based object detection model to replace RGB cameras, which may expose personal information. Their study applied transfer learning and achieved high occupancy detection performance even in low-resolution thermal imaging environments [4]. Kwon et al. proposed a method for predicting the number of occupants using inter-frame temperature changes and object tracking information from low-resolution thermal images. Their study estimated the number of indoor occupants by detecting people and tracking their movement paths using MSE, RGA, DBSCAN, and SORT algorithms [5].

Park et al. proposed an LSTM-based model that classifies occupant activities such as going out, staying indoors, cooking, cleaning, exercising, and sleeping using multi-sensor data composed of IAQ sensors, IR-UWB radar, and piezo sensors [6]. These studies demonstrate that various sensors can be used to analyze occupancy states and indoor environmental information. However, most studies focus on occupancy detection, occupant-count prediction, or activity classification, and thus have limitations in representing the semantic relationships among sensor data and state assessment results as an integrated context.

2.3 Ontology-Based Context and Rule-Based State Assessment Studies

Ontology is a knowledge representation method that supports semantic information representation and context-aware analysis by structurally defining the concepts, attributes, and relationships of a specific domain. In indoor environmental monitoring, it can be used to semantically connect entities such as Room, Sensor, Measurement, Occupancy, and EnvironmentState.

Meanwhile, rule-based state assessment determines results according to predefined thresholds and conditions, and therefore has the advantage that the assessment process is explicit and reproducible. Seneviratne et al. proposed a framework that provides trace-based, context-based, contrastive, and counterfactual explanations to verify inference results derived from rule-based knowledge systems and improve rule quality [7]. Their study shows that explaining the rule application process and the grounds for assessment to users can improve the transparency and verifiability of rule-based systems.

However, that study focuses on debugging existing rules and evaluating their quality, and it does not address a structure that converts real-time environmental sensor data into ontology-based context or directly links rule-based assessment results with LLM-based natural language explanation generation.

2.4 Hallucination Mitigation and Controlled Generation Structures

Generative language models can produce natural sentences, but they may cause hallucination by generating content that does not match the input information or cannot be verified from the input. Ji et al. classified hallucinations in natural language generation into intrinsic hallucinations, which contradict the input content, and extrinsic hallucinations, which cannot be verified from the input, and summarized various approaches for measuring and mitigating them [8].

Lewis et al. proposed a retrieval-augmented generation (RAG) structure that retrieves external documents and includes them in the generation input, rather than relying solely on knowledge stored in the parameters of a language model [9]. By using retrieved evidence documents in the generation process, RAG can improve the specificity and factuality of generated outputs for knowledge-intensive queries. However, RAG, which focuses on external document retrieval, has limitations when directly applied to indoor environmental monitoring, where immediate states must be assessed based on real-time sensor measurements and predefined environmental thresholds.

Yu et al. proposed a judgment-generation separation structure for immersive convergence-based fire safety education, in which an ontology and a rule-based judgment execution layer are constructed and generative AI is allowed to generate explanations only after the judgment results have been finalized [10]. That study is similar to the present study in that it structurally restricts generative AI from modifying rule-based judgments or arbitrarily proposing new actions.

However, previous study [10] primarily targets safety response judgment in fire situations and immersive convergence education, and it does not address the collection and integration of real-time indoor environmental sensor data, environmental state classification by space, occupancy-based HVAC linkage, or the provision of environmental information through a dashboard and TTS. This study is differentiated by integrating CO₂, PM_{2.5}, temperature, humidity, and occupancy data into an ontology-based context and restricting the LLM to explaining only the results of rule-based state assessment.

3. Controlled LLM Framework

3.1 Framework Architecture

This study proposes a controlled LLM framework for indoor environmental monitoring. The proposed structure aims to transform heterogeneous sensor data into ontology-based context and generate natural language explanations using the results of rule-based state assessment.

In the data collection layer, CO₂, PM_{2.5}, temperature, humidity, and occupancy data are collected using the EDC, IRC, and RDC.

Figure 1 shows the overall system architecture. The collected data are stored in MariaDB through the Edge Hub and are used for LLM-based explanation generation after ontology-based context construction and rule-based state assessment. The final results are provided to users through a React-based dashboard and TTS functionality.

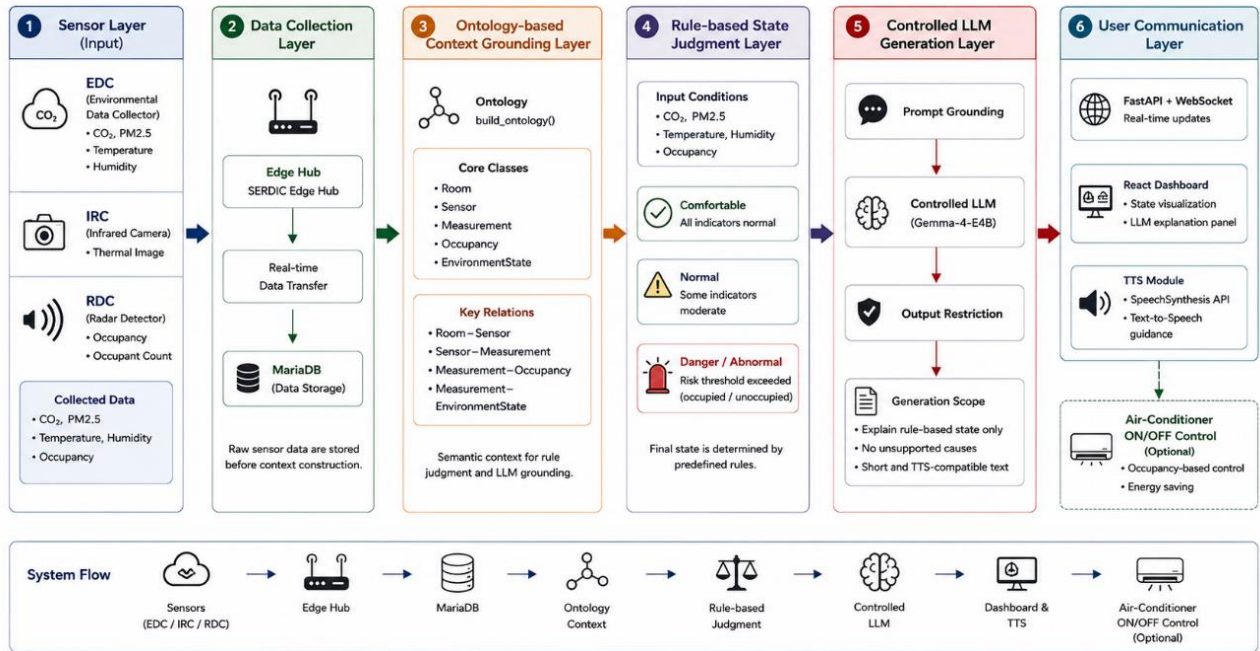


Figure 1. Architecture of the controlled LLM framework

3.2 Ontology-Based Context Structure

This study applied an ontology-based context structure to semantically connect heterogeneous sensor data and spatial information. The ontology consists of five classes: Room, Sensor, Measurement, Occupancy, and EnvironmentState.

Room represents spatial information, Sensor represents sensor devices, Measurement represents environmental measurements, Occupancy represents occupancy status, and EnvironmentState represents the result of rule-based state assessment. The EDC provides CO₂, PM2.5, temperature, and humidity data; the IRC supports thermal-image-based overheating detection and occupancy assessment; and the RDC is used as the primary sensor for determining occupancy.

Figure 2 shows the ontology-based context structure composed of the classes and their relationships.

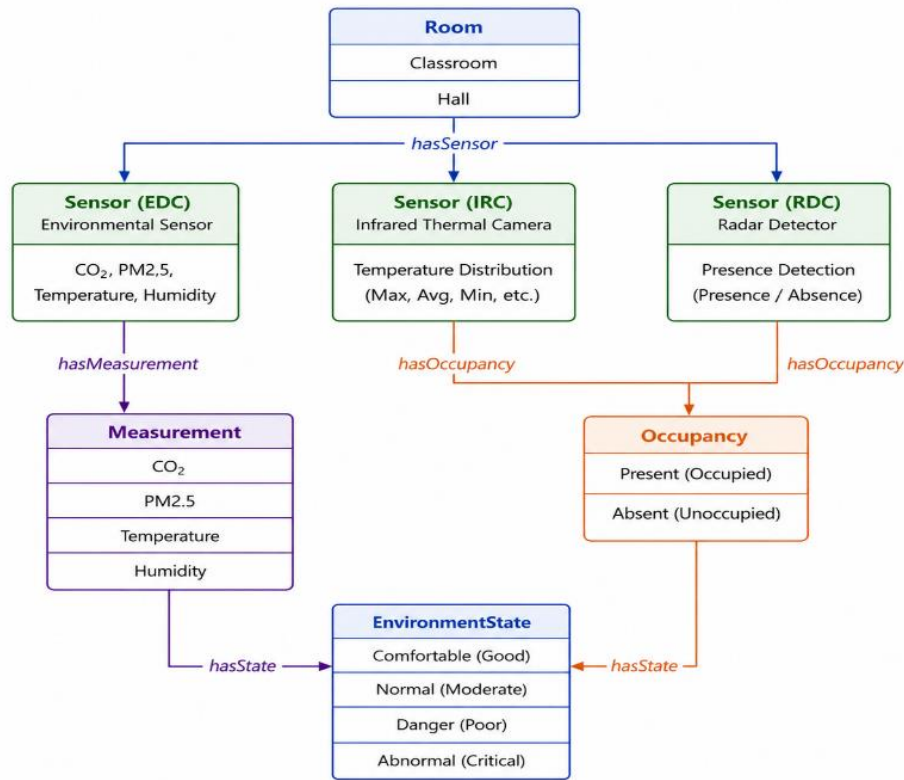


Figure 2. Ontology-based context structure

3.3 LLM-Based Environmental Interpretation

This study designed an LLM-based environmental interpretation structure using rule-based state assessment results and ontology-based context information, and applied prompt grounding to restrict the input scope.

4. Context Grounding and Rule-Based State Assessment

4.1 Ontology-Based Context Design

In this study, context was constructed by mapping spatial information and heterogeneous sensor data to ontology classes. Space, sensors, environmental measurements, and occupancy status are connected to Room, Sensor, Measurement, and Occupancy, respectively, while the results of rule-based state assessment are represented as EnvironmentState. The context constructed in this way is used for rule-based state assessment and LLM input.

4.2 Conversion of Sensor Data into Context

In this study, the `build_ontology()` function was implemented to integrate the latest sensor data stored in MariaDB by space and convert them into ontology-based context. For spaces in which multiple sensors were installed, representative values were generated using average and maximum values. Occupancy was determined by giving priority to the RDC, while the IRC was used as auxiliary information.

Occupancy was primarily determined using RDC data. When RDC data were available, the occupancy status derived from the RDC was used as the final occupancy information. When RDC data were not collected

or were unavailable, occupancy was secondarily estimated using the average temperature (stats_avg) obtained from the IRC. In cases where occupancy could not be determined using either the RDC or the IRC, the occupancy information was retained as None to handle missing-data conditions.

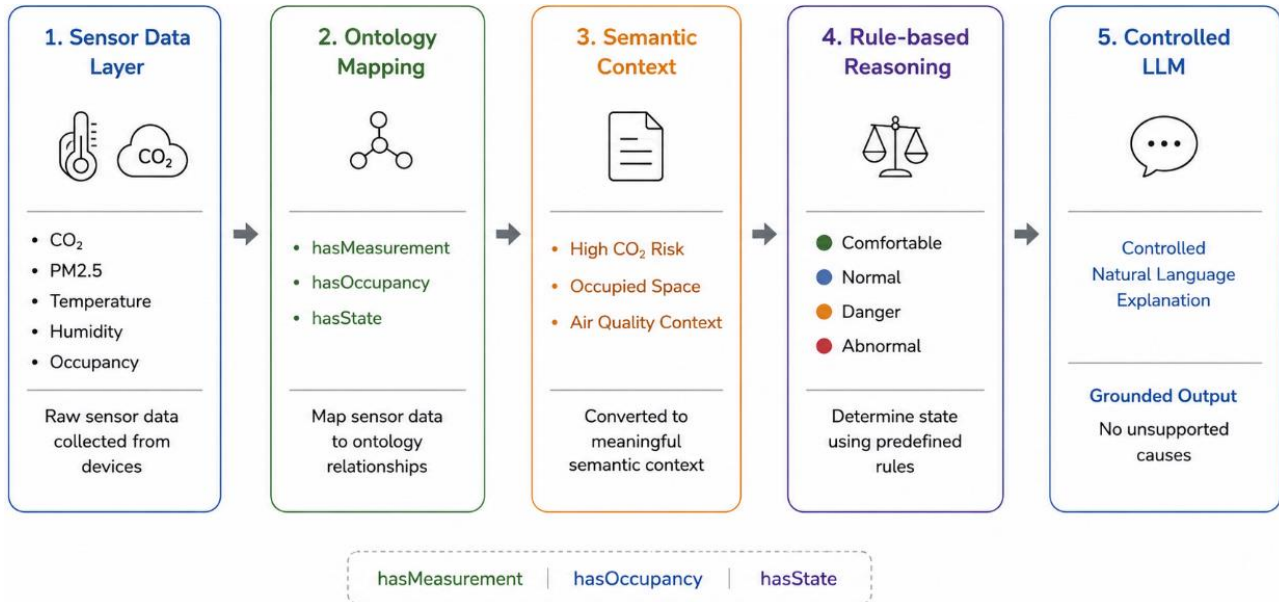


Figure 3. Context conversion process for sensor data

Figure 3 illustrates the process by which sensor data are converted into an ontology-based context structure. Raw sensor data are separated into measurement values and occupancy information and connected to Measurement and Occupancy, respectively. They are then finalized as EnvironmentState through rule-based state assessment.

Table 1 presents an example of sensor data context conversion and shows the process by which raw data are transformed into ontology relationships and semantic context. Through this conversion process, simple numerical data are extended into interpretable context information, which is then used as key data for rule-based state assessment and LLM input.

Table 1. Example of context conversion for sensor data

Item	Raw Data	Ontology Representation	Context
Space	room_id: 3F-LEFT	Room	Classroom
CO ₂	co2: 1050	hasMeasurement	High CO ₂
PM2.5	aerosol: 28	hasMeasurement	PM2.5 Level
Occupancy	target_count: 4	hasOccupancy	Occupied
Environmental state	danger	hasState	Danger State

The build_ontology() function collects the latest sensor data for each space and transforms them into a contextual structure corresponding to Room, Sensor, Measurement, Occupancy, and EnvironmentState. For example, sensor data from the 3F-LEFT space can be represented as an ontology-based context in the following JSON format.

```

{
  "room_id": "3F-LEFT",
  "type": "classroom",
  "hasSensor": {
    "EDC": ["EDC device IDs"],
    "RDC": ["RDC device IDs"],
    "IRC": ["IRC device IDs"]
  },
  "hasMeasurement": {
    "co2": 1050,
    "aerosol": 28,
    "temp": 26.0,
    "hum": 45.0,
    "stats_max": 32.5,
    "stats_avg": 29.4
  },
  "hasOccupancy": true,
  "occupantCount": 4,
  "hasState": "Danger"
}

```

The Context expressions in Table 1 were written in English as examples for conceptual explanation, whereas Korean-based expressions were used in the actual system implementation to improve user intuitiveness. In particular, the explanations or reasons generated from rule-based state assessment results were designed to be produced in Korean, such as “CO2 risk,” “occupancy state,” and “immediate action required.”

4.3 Rule-Based State Assessment

This study applied a rule-based state assessment structure. State assessment is performed by comprehensively considering CO2, PM2.5, temperature, humidity, and occupancy.

Environmental thresholds were set with reference to the Enforcement Rules of the School Health Act and the Indoor Air Quality Control Act [11]. A common criterion was applied to CO2, whereas different criteria were applied to PM2.5 depending on the type of space.

Table 2 presents the environmental thresholds by space type. These criteria are used to classify indoor environmental conditions into stages and serve as input conditions for rule-based state assessment.

In addition, the thermal-image-based sensor (IRC) was used as auxiliary information for occupancy assessment, and occupancy was considered possible when the average temperature (stats_avg) exceeded a certain criterion. The system was also configured to detect localized overheating based on the maximum temperature (stats_max).

Furthermore, when multiple environmental indicators simultaneously exceeded the normal range, compound assessment logic was applied to escalate the state, enabling more conservative risk assessment.

Table 2. Environmental thresholds by space type

Item	Comfortable	Normal	Danger
CO ₂	≤700 ppm	700~1000 ppm	>1000 ppm
PM2.5(classroom)	≤15	15~35	>35
PM2.5(central hall)	≤25	25~50	>50
Temperature	18~28°C	-	Outside range
Humidity	30~80%	-	Outside range

State assessment jointly considers occupancy and compound conditions. For example, even if the CO₂ concentration exceeds the danger threshold, the state is classified as abnormal if the space is unoccupied.

Table 3 presents the rule-based state classification criteria. These criteria are used as rules for determining the final state by jointly reflecting the environmental state and occupancy.

Table 3. State classification criteria

State	Assessment Condition
Comfortable	All indicators are within the normal criterion range
Normal	Some indicators fall within the normal range and no danger condition is present
Danger	Danger threshold exceeded or multiple normal-level indicators occur + occupied
Abnormal	Danger threshold exceeded + unoccupied

To improve the reproducibility of state assessment, the final state was configured to be determined according to the following rules.

First, if neither CO₂ nor PM2.5 data are collected, the condition is regarded as the absence of sensor data and classified as Abnormal.

Second, if all environmental indicators satisfy the comfortable criteria, the state is classified as Comfortable.

Third, if one or more environmental indicators fall outside the comfortable criteria but do not exceed the danger thresholds, the state is classified as Normal.

Fourth, if either CO₂ or PM2.5 exceeds the danger threshold, the state is classified as Danger.

Fifth, if two or more environmental indicators simultaneously fall outside their normal criterion ranges, the condition is regarded as a compound environmental anomaly, and the state is escalated to Danger.

5. Environmental Interpretation Based on a Controlled LLM

5.1 Structure of Controlled LLM-Based Environmental Interpretation

This study designed a controlled LLM-based environmental interpretation structure using rule-based state

assessment results and ontology-based context. The final state is determined in the rule-based layer, and the LLM is restricted to performing only the role of explaining the result in natural language.

As shown in Figure 4, sensor data and state assessment results are combined during the prompt grounding stage, and the controlled LLM generates explanations only within a restricted scope.

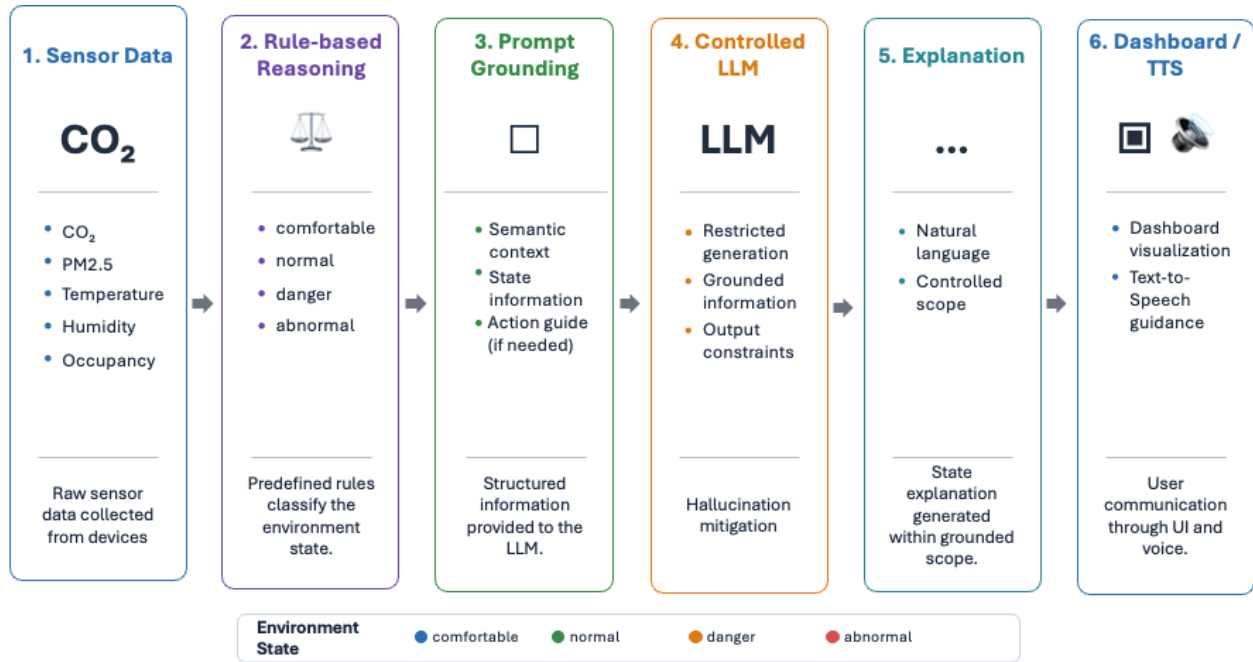


Figure 4. Structure of controlled LLM-based environmental interpretation

5.2 Output Restriction Based on Prompt Grounding

Prompt grounding is a method that structures sensor values, space type, occupancy, and environmental state and provides them as input to the LLM. This restricts the LLM from generating causal inferences or excessive explanations beyond the scope of the input.

Table 4 shows the differences in input structures among the general LLM, ontology-based LLM, and controlled LLM, while Table 5 summarizes the output restriction strategies applied in this study.

Table 4. Example comparison of LLM inputs

Category	Prompt Example
General LLM	CO2=1050, PM2.5=28, Temp=26
Ontology-based LLM	Classroom, CO2 HIGH, PM2.5 Level, Occupied
Controlled LLM	Classroom, Occupied, Danger State, Ventilation Recommended

Table 5 presents the controlled LLM strategies applied in this study. Each control strategy is applied to restrict the output scope of the LLM and maintain the consistency of state-based explanations.

Table 5. Controlled LLM strategies

Control Method	Purpose
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Rule-based grounding	Hallucination mitigation
Ontology-based context injection	Maintaining semantic consistency
Output length restriction	Ensuring TTS compatibility
Markdown restriction	Maintaining natural language output
State-based prompt construction	Generating state-consistent explanations

Rule-based grounding and ontology-based context injection improve the consistency and reliability of explanations by restricting the system to generate explanations only within the scope of the state assessment results and semantic context, thereby effectively reducing hallucinations such as unnecessary inference or the generation of external information.

5.3 Natural Language Explanation Generation and TTS Integration

The generated explanations are delivered in real time to the React dashboard through FastAPI and WebSocket and are also used for TTS guidance through the SpeechSynthesis API. To ensure TTS compatibility, the output was limited to natural language within two sentences, and the use of Markdown symbols, lists, and special characters was restricted.

Table 6 presents examples of natural language explanations generated for each state. These examples show how explanations generated according to rule-based state assessment results are expressed and confirm that concise explanations consistent with the corresponding states can be generated.

Table 6. Examples of natural language explanations by state

State	Generated Explanation
Comfortable	The current indoor environment is in a comfortable state.
Normal	Some environmental values are increasing.
Danger	The current CO2 concentration is high, and ventilation is required.
Abnormal	The sensor status should be checked, or the space should be inspected.

This study reduced the hallucination problem that may occur in conventional LLM-based environmental interpretation through a controlled LLM-based natural language explanation generation structure and enabled stable generation of explanations consistent with state assessment results. This confirms that reliable environmental information can be provided to users in an intuitive manner.

6. Experiments and Results

6.1 Experimental Environment

This study conducted experiments in classrooms and central halls on two floors of a building at K University. EDC, IRC, and RDC sensors were installed to collect indoor environmental data.

Figure 5 shows the sensor deployment structure in the experimental environment. The sensor configuration and placement were applied differently depending on the space type, and the deployment was designed to enable environmental data collection and occupancy information acquisition.

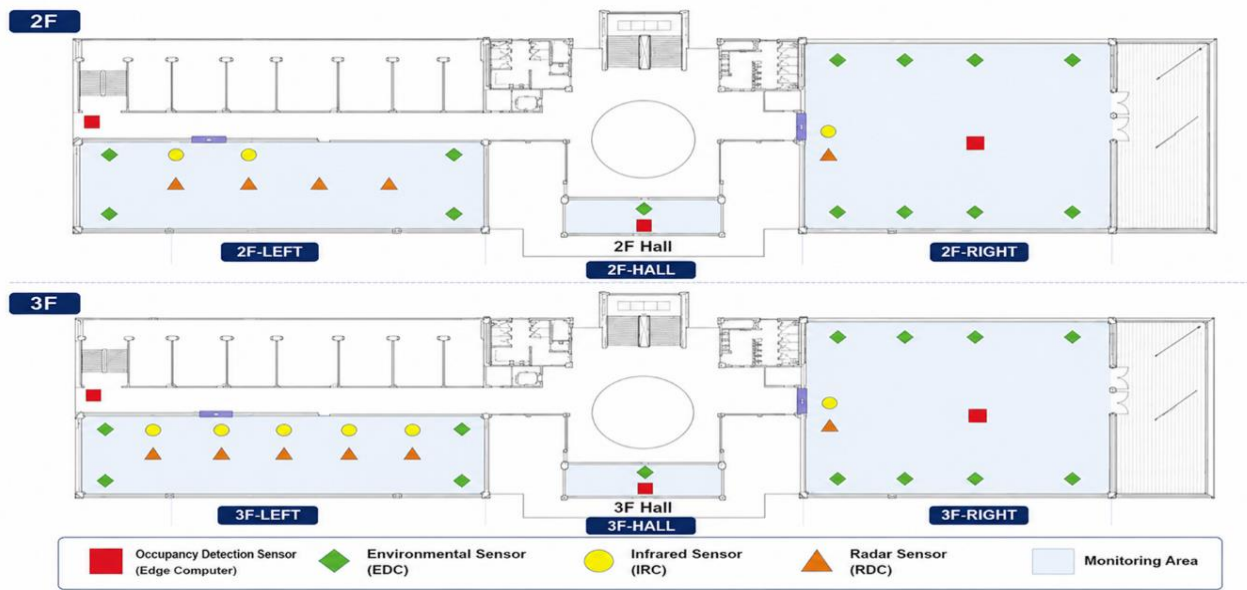


Figure 5. Sensor deployment structure in indoor spaces

Table 7 presents the server and software environment used in this study. This environment includes system components for LLM-based environmental interpretation and real-time data processing.

Table 7. Experimental environment

Item	Description
Server model	TX2550 M7
RAM / GPU	128GB / 48GB
Operating system	Ubuntu 24.04.2 LTS
LLM inference engine	Ollama
LLM model	Gemma4-E4B
Database	MariaDB
Backend	FastAPI
Frontend	React

Table 8 presents the sensor configuration by space. The number and type of sensors were applied differently depending on the space type. In particular, the central halls adopted a simplified sensor structure centered on a single EDC sensor for measuring environmental conditions.

Table 8. Sensor configuration by space

Space ID	Space Type	EDC	IRC	RDC
2F-LEFT	Classroom	4	2	4
2F-HALL	Central hall	1	-	-
2F-RIGHT	Large classroom	8	1	1
3F-LEFT	Classroom	4	5	5
3F-HALL	Central hall	1	-	-
3F-RIGHT	Large classroom	8	1	1

This study used indoor environmental sensor data collected in real time from two floors of a building at K University. During the experimental period, 4,848,334 environmental sensor data records, 8,564,517 radar sensor data records, and 37,207,397 thermal imaging sensor data records were collected. The collected data were used for ontology-based semantic context generation and rule-based state assessment.

6.2 Evaluation Metrics

This study used the State Consistency Rate (SCR) and Hallucination Rate (HR) as evaluation metrics to assess the reliability and consistency of LLM-based environmental explanations.

The State Consistency Rate (SCR) refers to the degree of agreement between the rule-based state assessment results and the LLM explanation results and is defined as follows.

$$SCR = \frac{N_{\{matched\}}}{N_{\{total\}}} \quad (1)$$

Here, $N_{\{matched\}}$ denotes the number of explanations that match the rule-based state assessment results, and $N_{\{total\}}$ denotes the total number of evaluation samples.

The Hallucination Rate (HR) refers to the proportion of generated content that is not included in the input data or state assessment results and is defined as follows.

$$HR = \frac{N_{\{hallucination\}}}{N_{\{total\}}} \quad (2)$$

Here, $N_{\{hallucination\}}$ denotes the number of explanations in which hallucination occurred.

6.3 Results of Rule-Based State Assessment

This study performed rule-based state assessment based on the collected sensor data. State assessment was conducted by comprehensively considering CO₂, PM_{2.5}, temperature, humidity, and occupancy and classifying the state into comfortable, normal, danger, or abnormal.

Table 9 presents examples of state assessment results based on actual sensor data. These results show that state classification may differ depending on occupancy even under the same environmental measurement conditions.

Table 9. Examples of state assessment results

Space	CO ₂ (ppm)	PM _{2.5}	Temperature (°C)	Occupancy	State
3F-LEFT	438	5	24.8	Occupied	comfortable
3F-LEFT	619	5	25.8	Occupied	normal
3F-LEFT	1150	5	26.6	Occupied	danger
3F-LEFT	1131	5	26.8	Unoccupied	abnormal

As shown in Table 9, even when the same CO₂ concentration exceeded the danger threshold, the state was classified as either danger or abnormal depending on occupancy. This shows that the proposed structure classifies environmental states by reflecting not only thresholds but also occupancy information and semantic context.

6.4 Comparison of Controlled LLM Outputs

This study compared three structures: a general LLM, an ontology-based LLM, and a controlled LLM. The general LLM directly receives sensor values as input, while the ontology-based LLM adds semantic context; however, both structures rely on the LLM for final state assessment. In contrast, the controlled LLM is restricted to explaining only the results of rule-based assessment.

The experiment used 30 actual sensor data samples, and the same Gemma4 model was applied to all three structures.

Table 10. Comparison of outputs from the general LLM and the controlled LLM

Category	General LLM	Ontology-Based LLM	Controlled LLM
Input structure	Direct input of sensor values	Includes context information	Includes context + state results
State assessment	LLM-dependent	LLM-dependent	Rule-based assessment
State explanation	Includes causal speculation	Partially reflects state	State-based explanation
Action guidance	May provide excessive guidance	Partially restricted	Restricted guidance
Hallucination occurrence	High	Moderate	Low
Explanation consistency	Low	Moderate	High
State consistency	Partial match	Partial improvement	High match

As shown in Table 10, the general LLM and ontology-based LLM showed limitations in state consistency and output consistency because their final state assessment depended on the LLM. In contrast, the controlled LLM was restricted to explaining only the results of rule-based assessment, thereby reducing unnecessary causal inference and stably generating explanations consistent with the state.

6.5 Experimental Results and Analysis

SCR was measured based on the semantic agreement of state keywords between the rule-based state assessment results and the LLM output text. If keywords with meanings opposite to the state were included simultaneously, the result was judged as inconsistent. HR was measured through manual review and keyword analysis based on whether causal speculation, excessive action guidance, or external situational inference not included in the input data or state assessment results occurred.

Table 11. Quantitative evaluation results by LLM structure

Model	SCR	HR
LLM Only	40.0%	86.7%
LLM + Ontology	13.3%	66.7%
Controlled LLM	96.7%	0.0%

The experimental results show that the controlled LLM achieved a state consistency rate of 96.7% and a

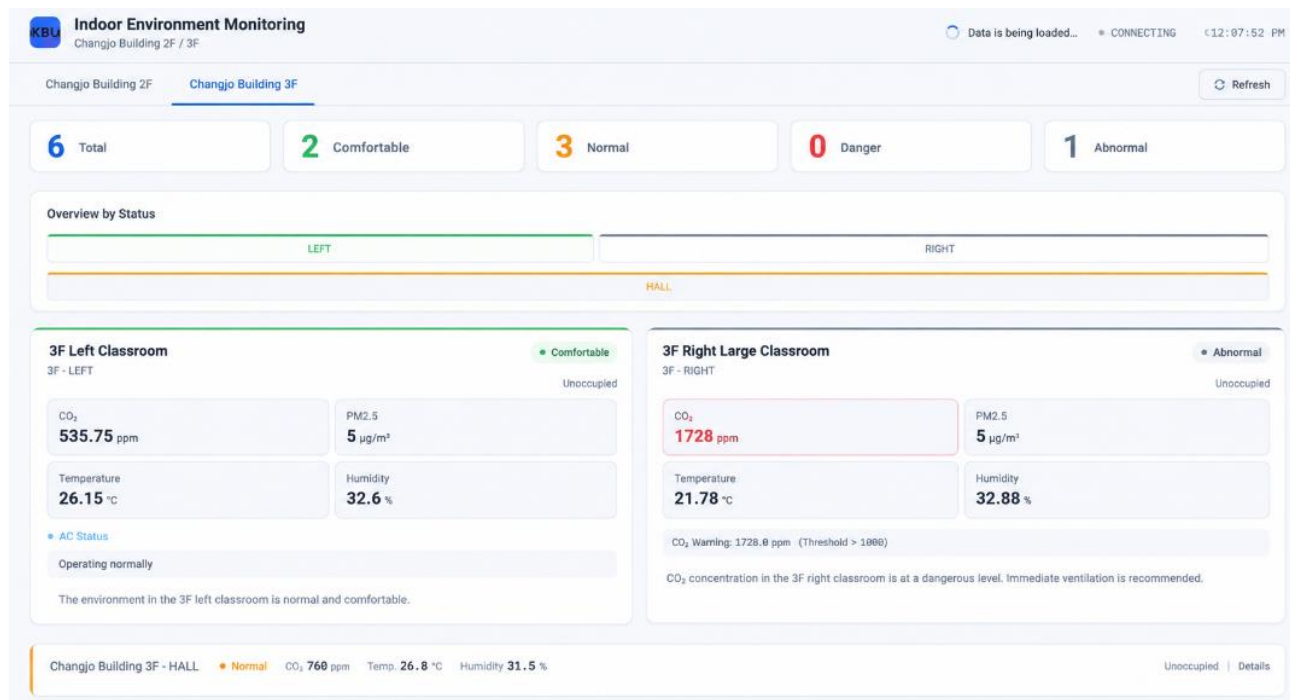
hallucination rate of 0.0%. This performance is higher than that of the LLM Only and LLM + Ontology structures. This improvement is considered to result from the joint use of rule-based state assessment results and ontology-based semantic context, which enhanced the consistency of state explanations.

The general LLM exhibited problems such as generating causal inferences or excessive risk expressions that were not related to the actual sensor state. The ontology-based LLM was able to semantically structure the input data; however, because final state assessment still depended on the LLM, it had limitations in improving explanation consistency. In contrast, the controlled LLM was restricted to explaining only the results of rule-based assessment, thereby reducing unnecessary inference and stably generating explanations consistent with the actual state.

6.6 Real-Time Dashboard and TTS Results

In this study, the generated natural language explanations were delivered to users by linking them with a real-time dashboard and a TTS voice guidance function [12]. The generated results are transmitted to the user interface in real time through a FastAPI-based WebSocket and are provided as voice guidance using the SpeechSynthesis API upon user request.

Figure 6 shows the implemented real-time environmental monitoring dashboard. Figure 6(a) shows a screen that allows users to intuitively check the status of all spaces, while Figure 6(b) shows a panel that provides environmental explanations and TTS-based voice guidance.



(a) Overall space status monitoring screen



(b) Environmental explanation and TTS guidance panel

Figure 6. Real-time environmental monitoring dashboard screen

7. Conclusion and Future Work

7.1 Conclusion

This study proposed a controlled LLM framework that converts indoor environmental data collected by EDC, IRC, and RDC into ontology-based context and restricts the LLM to explaining only the results of rule-based state assessment. The proposed structure classifies indoor environmental states into comfortable, normal, danger, and abnormal and generates natural language explanations based on these results.

In addition, this study aimed to reduce the hallucination problem that may occur in conventional LLM-based environmental interpretation by restricting LLM outputs using rule-based state assessment results and semantic context as grounding information. In particular, the LLM was restricted from independently performing state assessment and was limited to serving as an auxiliary component that explains the results of rule-based state assessment.

The experimental results confirmed that the proposed structure improved state consistency and explanatory reliability compared with a general LLM and reduced causal speculation and excessive action guidance. The generated explanations were linked with a WebSocket-based real-time dashboard and TTS voice guidance function, and an air-conditioner power ON/OFF control function based on occupancy was also implemented. In future work, the energy management function will be expanded by linking ventilation equipment, cooling modes, set-point temperature control, and airflow control.

The main contributions of this study are as follows.

First, heterogeneous indoor environmental sensor data were integrated into an ontology-based semantic context structure.

Second, explainability and reliability were improved by combining rule-based state assessment with a controlled LLM structure.

Third, the hallucination problem of the LLM was reduced by applying a context grounding structure.

Fourth, by linking the proposed system with a real-time dashboard and TTS-based voice guidance function, this study demonstrated its potential as a safety education support system.

7.2 Future Work

Although this study confirmed the feasibility of a controlled LLM framework for indoor environmental monitoring, several limitations remain. Because this study was conducted in a single building environment, there are limitations in generalizing the results to diverse spatial environments, and some evaluations were conducted based on manual assessment.

First, the current state assessment is performed mainly using a rule-based threshold structure and therefore does not sufficiently reflect complex environmental change patterns or long-term trends. Future work should add environmental change prediction functionality by integrating time-series-based AI models.

Second, because the current ontology structure is centered on indoor environmental data, it does not sufficiently reflect diverse contextual information such as user behavior patterns, energy usage information, and external environmental information. Future work should expand the scope of semantic context through knowledge graph-based structural extension.

Third, although this study used a controlled LLM structure with rule-based grounding, the evaluation was conducted primarily using a specific model, Gemma4, and sufficient research has not been conducted on performance comparisons among diverse LLM models or prompt optimization. Therefore, future work should compare and analyze explanatory reliability and hallucination reduction effects across Gemma, Llama, and GPT-family models.

Fourth, although the current system is centered on indoor environmental monitoring and safety education support, it can be extended to various domains such as smart campuses, smart buildings, and smart healthcare environments. In particular, when linked with a multi-agent-based environmental control system, it has the potential to evolve into a system for automatic ventilation control, energy optimization, and risk situation response.

In future work, this research will be extended into a real-time environmental intelligence system based on explainable artificial intelligence (Explainable AI) and context-aware intelligence by building on the ontology-based context structure and controlled LLM structure.

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